

Announcements – Jan 29

- HW 2 is due Friday at 9am.
- SI sessions are Tue 6-7pm and Thu 2-3pm.
- Quiz 6 trivia – The first arcade video game was "Computer War" (1971). It was based on a mainframe game called "Space War" which many believe was the first video game. Basically, it was a two-player game where two spaceships flew around the screen shooting at each other. 7% of you got this correct. The most common answer was Centipede.

Prenex Normal Form (PNF)

For every quantified statement, there is an equivalent quantified statement that is in PNF.

Casually: “Every quantified statement can be written in PNF.”

How?

If every quantifier has a different variable, then just push them all to the left without crossing any of the quantifiers.

$$\forall x (P(x) \rightarrow \exists y D(y, x))$$

$$\forall x \exists y (P(x) \rightarrow D(y, x))$$

Prenex Normal Form (PNF)

For every quantified statement, there is an equivalent quantified statement that is in PNF.

Casually: “Every quantified statement can be written in PNF.”

What do we do if some names are the same?

We first change the variables names so that no quantifier has the same variable name

$$\exists x P(x) \wedge \exists x Q(x)$$

$$\exists x P(x) \wedge \exists y Q(y)$$

$$\exists x \exists y (P(x) \wedge Q(y))$$

Prenex Normal Form (PNF)

There is a kind of weird case with implication and quantifiers.

Consider: $\exists x P(x) \rightarrow \exists y Q(y) \equiv (\exists x P(x)) \rightarrow \exists y Q(y)$

which means "if there is an x which makes $P(x)$ true, then there is a y which makes $Q(y)$ true"

This is not the same as $\exists x \exists y (P(x) \rightarrow \exists y Q(y))$

which means "there is an x and a y such that $P(x)$ implies $Q(y)$ ".

Notice this will be true if any x exists that makes $P(x)$ false.

So what's going on, and how do we write $\exists x P(x) \rightarrow \exists y Q(y)$ in PNF?

Prenex Normal Form (PNF)

$$\exists x P(x) \rightarrow \exists y Q(y) \equiv \neg(\exists x P(x)) \vee \exists y Q(y)$$

$$\equiv \forall x (\neg P(x)) \vee \exists y Q(y)$$

$$\equiv \forall x \exists y (\neg P(x) \vee Q(y))$$

$$\equiv \forall x \exists y (P(x) \rightarrow Q(y))$$

which means for every x , there exists a y , so that if $P(x)$ is true, then $Q(y)$ is true.

So, if a quantifier is bound to the hypothesis of a implication, and not to the entire implication, it is actually behind a negation.

End of Section 1.5
Starting Section 1.6

Rules of Inference

- An *argument* is a sequence of statements (propositions) called the *hypotheses* (or *premises*), followed by a statement (proposition) called the *conclusion*.
- An argument is called *valid* if all the propositions being true implies the conclusion must be true.

Valid Argument

p_1

p_2

...

p_n

$\therefore C$

<- read "therefore c"

$(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow c$ is a tautology

Valid Argument

p_1

p_2

...

p_n

$\therefore C$

We can show an argument is valid via a truth table:

We just need to make sure c is true on all rows

where $p_1, p_2, \dots, p_n$ are all true

Valid Argument

We can show this argument is valid using a truth table:

Example:

$p \rightarrow q$

$q \rightarrow r$

$\therefore p \rightarrow r$

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Valid Argument

We only care about the rows where everything in the hypothesis is true

Example:

$p \rightarrow q$

$q \rightarrow r$

$\therefore p \rightarrow r$



p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Valid Argument

If the conclusion is true in each of those rows, the argument is valid

Example:

$p \rightarrow q$

$q \rightarrow r$

$\therefore p \rightarrow r$



p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Valid Argument

To show an argument is not valid, you just need to show one case where all propositions in the hypothesis are true, but the conclusion is not.

Example:

$p \rightarrow r$
 $q \rightarrow r$

 $\therefore p \rightarrow q$

Let:
 $p = T$
 $q = F$
 $r = T$

Then:
 $p \rightarrow r$ is T
 $q \rightarrow r$ is T
but $p \rightarrow q$ is F

So the
argument is not
valid!

Classic Philosophy Example

- All Cats have four legs
- Mindy is a cat.
- Therefore, Mindy has four legs.

Classical, valid argument in English

Another Example

- f: You can fly
- s: You are Superman

Argument:

$s \rightarrow f$

s

$\therefore f$

Is it valid?

If you are Superman, then you can fly.
You are Superman. Therefore, you can fly.

Another Example

- f: You can fly
- s: You are Superman

Argument:

$s \rightarrow f$

s

$\therefore f$

Is it valid?

yes because

$(s \wedge (s \rightarrow f)) \rightarrow f$ is a tautology

If you are Superman, then you can fly.
You are Superman. Therefore, you can fly.

Valid vs Sound

- We've seen the definition for a valid arguments.
 - p, q, r, s, c is *valid* iff $p \wedge q \wedge r \wedge s \rightarrow c$ is a tautology
- An argument is *sound* iff it is valid and all the premises are true
- For example: If I am Superman then I can fly. I can fly. Therefore, I am Superman. Is NOT valid (or sound)
- If I have a fedora, then I am Perry the Platypus. I have a fedora. Therefore, I am Perry the Platypus. Is valid, but not sound.

Rules of Inference

- Some established, relatively simple valid argument forms are gathered together and called the *rules of inference*.

Rules of Inference

Name	premises	conclusion
Modus ponens	$p, p \rightarrow q$	q
Modus tollens	$\neg q, p \rightarrow q$	$\neg p$
Hypothetical syllogism	$p \rightarrow q, q \rightarrow r$	$p \rightarrow r$
Disjunctive syllogism	$p \vee q, \neg p$	q
Addition	p	$p \vee q$
Simplification	$p \wedge q$	p
Conjunction	p, q	$p \wedge q$
Resolution	$p \vee q, \neg p \vee r$	$q \vee r$

Rules of Inference Examples

- If I am superman, then I have x-ray vision. If I have x-ray vision I can see who's outside. Therefore, if I am Superman, I can see who's outside

s: I am Superman

x: I have x-ray vision

o: I can see who is outside

$s \rightarrow x$

$x \rightarrow o$

$\therefore s \rightarrow o$

Rules of Inference Examples

- If I am superman, then I have x-ray vision. If I have x-ray vision I can see who's outside. Therefore, if I am Superman, I can see who's outside

s: I am Superman

x: I have x-ray vision

o: I can see who is outside

$s \rightarrow x$

$x \rightarrow o$

$\therefore s \rightarrow o$

Hypothetical syllogism

Rules of Inference Examples

- If I am superman, then I'm faster than a speeding bullet. I am Superman.
Therefore, I am faster than a speeding bullet.

s: I am Superman

f: I am faster than a speeding bullet

$s \rightarrow f$

s

$\therefore f$

Rules of Inference Examples

- If I am superman, then I'm faster than a speeding bullet. I am Superman.
Therefore, I am faster than a speeding bullet.

s: I am Superman

f: I am faster than a speeding bullet

$s \rightarrow f$

s

$\therefore f$

Modus ponens

Rules of Inference Examples

- Either Superman will defeat Batman or Batman will have kryptonite. Batman defeats Superman. Therefore, Batman has kryptonite.

s: Superman defeats Batman

k: Batman has kryptonite

$s \vee k$

$\neg s$

$\therefore k$

Rules of Inference Examples

- Either Superman will defeat Batman or Batman will have kryptonite. Batman defeats Superman. Therefore, Batman has kryptonite.

s: Superman defeats Batman

k: Batman has kryptonite

$s \vee k$

$\neg s$

$\therefore k$

Disjunctive syllogism

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.
- Can we conclude that, if I don't send you flowers, I'll do well on the quiz?

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.
- Can we conclude that, if I don't send you flowers, I'll do well on the quiz?

Begin by identifying the propositions:

p: You email me a love note

q: I send you flowers

r: I study Discrete Math

s: I do well on the quiz

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.
- Can we conclude that, if I don't send you flowers, I'll do well on the quiz?

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

p: You email me a love note

q: I send you flowers

r: I study Discrete Math

s: I do well on the quiz

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

1. $p \rightarrow q$

hypothesis

$p \rightarrow q$

$\neg p \rightarrow r$

$r \rightarrow s$

Conclusion:

$\therefore \neg q \rightarrow s$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

1. $p \rightarrow q$

hypothesis

2. $\neg q \rightarrow \neg p$

contrapositive (1)

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

1. $p \rightarrow q$

hypothesis

2. $\neg q \rightarrow \neg p$

contrapositive (1)

$$p \rightarrow q$$

3. $\neg p \rightarrow r$

hypothesis

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

1. $p \rightarrow q$

2. $\neg q \rightarrow \neg p$

3. $\neg p \rightarrow r$

4. $\neg q \rightarrow r$

hypothesis

contrapositive (1)

hypothesis

hypothetical syllogism (2, 3)

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

1. $p \rightarrow q$

2. $\neg q \rightarrow \neg p$

3. $\neg p \rightarrow r$

4. $\neg q \rightarrow r$

5. $r \rightarrow s$

hypothesis

contrapositive (1)

hypothesis

hypothetical syllogism (2, 3)

hypothesis

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

Conclusion:

$$\therefore \neg q \rightarrow s$$

1. $p \rightarrow q$

hypothesis

2. $\neg q \rightarrow \neg p$

contrapositive (1)

3. $\neg p \rightarrow r$

hypothesis

4. $\neg q \rightarrow r$

hypothetical syllogism (2, 3)

5. $r \rightarrow s$

hypothesis

6. $\neg q \rightarrow s$

hypothetical syllogism (4, 5)

This is called a logical proof

Rules of Inference with Quantifiers

- We now want to prove conclusions with quantified propositions.
- To apply the rules of inference in these cases we must remove the quantifier by plugging in a value from the domain for the variable.
- We may either plug in an **arbitrary** element of the domain that has no special properties, or a **particular** element that has some special properties.

Rules of Inference with Quantifiers

The domain is D

Universal Instantiation:

$\forall x P(x)$ $\therefore P(c)$ (for any arbitrary or particular $c \in D$)

Example:

D = all cats

Every cat has four legs.

Mindy is a cat.

Therefore, Mindy has four legs

Rules of Inference with Quantifiers

The domain is D

Universal Generalization:

c is an arbitrary element of D

$P(c)$

$\therefore \forall x P(x)$

Example:

D = all cats

Let c be an arbitrary cat

c has four legs

Therefore, every cat has four legs

Rules of Inference with Quantifiers

The domain is D

Existential Instantiation:

$\exists x P(x)$

$\therefore P(c)$ some particular $c \in D$

Example:

D = all cats

There is a cat with a grey nose

Mindy is a cat with a grey nose

Rules of Inference with Quantifiers

The domain is D

Existential Generalization:

$P(c)$

$\therefore \exists x P(x)$

some particular/arbitrary $c \in D$

Example:

$D =$ all cats

Mindy is a cat with a grey nose

There is a cat with a grey nose

Rules of Inference for Predicates

Example(s):

Everyone in this class has had a programming course.

Hugo is in this class. Has he had a programming course?

Rules of Inference for Predicates

Example(s):

Everyone in this class has had a programming course.
Hugo is in this class. Has he had a programming course?

Yes! Here's how we know:

$C(x)$: x is in this class, $x \in \text{People}$

$P(x)$: x has had a programming course, $x \in \text{People}$

- | | | |
|-------|---|--------------------------|
| (1) | $\forall x (C(x) \rightarrow P(x))$ | (Hypothesis) |
| (2) | $C(\text{Hugo})$ | (Hypothesis) |
| (3) | $C(\text{Hugo}) \rightarrow P(\text{Hugo})$ | (1, Univ. Instantiation) |
| <hr/> | | |
| (4) | $\therefore P(\text{Hugo})$ | (2, 3, Modus Ponens) |